KENYA
Forest Service

MANGROVE HARVEST MANAGEMENT PLAN FOR LAMU COUNTY (2025 - 2035)

KMFRI

KENYA FORESTRY RESEARCH INSTITUTE KEFRI

KENYA WILDLIFE SERVICE

Collaborating Community Forest Associations

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PLAN APPROVAL

MANGROVE HARVEST MANAGEMENT PLAN

FOR LAMU COUNTY

Prepared by Kenya Forest Service

For the 10-year period from 2025 – 2035

This Mangrove Harvest Management Plan for Lamu County is hereby approved for implementation. It will be guided by approved annual work plans. The Plan shall be amended as need arises, reviewed after five years, and revised after ten years from the date of approval.

Approved by: Date

_____ _____

Alex L. Lemarkoko EBS, 'ndc'(K)

Chief Conservator of Forests

Kenya Forest Service

PLAN RECOMMENDATION FOR APPROVAL

This Mangrove Harvest Management Plan has been prepared by a task force composed of staff from Kenya Forest Service, Kenya Marine and Fisheries Research Institute, Kenya Forestry Research Institute, County Government of Lamu, and Community Forest Associations with support from United Nations Environment Programme and The Nature Conservancy in line with recommendations from the National Mangrove Management Committee.

_____ _____

Forest Station Manager *Date*

_____ _____

County Forest Conservator *Date*

_____ _____

Regional Forest Conservator *Date*

_____ _____

Head of Forest Management Planning *Date*

_____ _____

Deputy Chief Conservator of Forests-

Natural Forest Conservation *Date*

Foreword

For many years, the management of mangrove resources in Kenya focused on the removal of mangrove wood products, overlooking their vital ecosystem services such as supporting fisheries, biodiversity conservation, climate regulation, and shoreline protection, including carbon sequestration and storage. To address this challenge, the National Mangrove Ecosystem Management Plan (2017–2027) was developed to provide a roadmap towards the rehabilitation, conservation, and sustainable utilization of mangrove resources in Kenya. The plan identifies the need to develop a sustainable Harvest management plan for the mangroves of Lamu County.

Commercial harvest of mangrove wood is a livelihood activity for many households in Lamu County. While KFS regulates this activity through a licensing system, the current practice is largely market-driven rather than resource-based. This has led to the over-exploitation of preferred pole sizes for construction in some areas.

This 10-year Harvest management plan for Lamu gives guidelines on sustainable harvesting of various pole sizes across all management blocks where commercial harvesting is permitted during the 2025-2035 period. Poles of all utilization classes will be selectively harvested from a single management block each year to promote uniformity in the forest structure over time. This approach ensures that, as the current *Fito* poles (are on average around 7 years old) mature into *Nguzo (*are obtained at 40 years), the stand will maintain a balanced distribution of tree sizes as all size classes will have been systematically harvested and left to regenerate.

Acknowledgements

A National Task Force commissioned by the Kenya Forest Service (KFS) comprising the Kenya Marine and Fisheries Research Institute (KMFRI), and the Kenya Forestry Research Institute (KEFRI) and KFS consisted a team of technical experts that also worked closely with the County Government of Lamu, Community Forest Associations (CFAs), community leaders and other key stakeholders.

The Service appreciates the work undertaken by all the resource persons in compilation of this Harvest Management Plan. KFS also acknowledges funding support from The Nature Conservancy (TNC) and the United Nations Environment Programme (UNEP).

Committee Members responsible for drafting the Lamu Mangrove Harvest Management Plan

Charity Munyasya, 'ndc'(K)	KFS
James Kairo, PhD	KMFRI
Stanley Nadir, PhD	KEFRI
James Mburu	KFS
JamesMwangombeMwamodenyi	KFS
Nasib Ali Mwamutsi	KFS

The following resource persons contributed to development of Lamu Mangrove Harvest Management Plan

Amina Juma Hamza, PhD	KMFRI
Anthony Mbatha	KMFRI
Chemuku Wekesa, PhD	KEFRI
David Ouma	TNC
Dominic Musango	KFS
Fredrick Mburu	KMFRI
George Maina	TNC
Henry M. Komu	KEFRI
Judith Okello, PhD	KMFRI
Kipkorir Sigi Lang'at, PhD	KMFRI
Kisilu Mary	KMFRI
Mohamed Abubakar	County Government of Lamu
Safi Ibrahim	KFS

Secretariat

Brian Githinji Kiiru KMFRI

Zipporah Muchiri KMFRI

Executive Summary

The Forest Conservation and Management Act (2016) mandates that all forests in Kenya be managed through approved management plans. The National Mangrove Ecosystem Management Plan (2017–2027) provides the overall framework for the sustainable management of mangrove forests.

Mangrove forests in Lamu cover about 37,350 ha, representing about 61% of the total mangrove area in the country. All the nine mangrove tree species described in Kenya are represented in Lamu, with *Rhizophora mucronata* (Mkoko) and *Ceriops tagal* (Mkandaa) being the most dominant. For their effective management, mangrove forests in Lamu County have been subdivided into five management blocks: Northern swamps, Northern Central swamps, Mongoni-Dodori Creek swamps, Pate Islands swamps, and Southern swamps. Northern Central swamps are the largest, accounting for 34% of the total mangroves in Lamu, followed by the Pate Island swamps (26%), Mongoni-Dodori Creek swamps (17%), Southern swamps (14%), and Northern swamps (8%).

Most commercial harvesting of mangroves is concentrated in the Northern Central, Mongoni-Dodori Creek, Pate Island, and Southern swamps. The harvested products are classified in diameter classes to facilitate effective mangrove wood removals. *Boriti*-sized poles with butt diameters from 9.1cm to 13.0cm are the most popular, followed by *Mazio* (6.1cm to 9.0cm), *Pau* (4.1cm to 6.0cm), *Fito* (less than4cm), and *Banaa* (20.1cm to 35cm).

Approximately 1,739 ha (4.87%) of mangrove forests in Lamu were lost between 1990 and 2019, translating to an annual loss of 60 ha (0.16%). The highest decline of 114 ha yr^{-1} was observed between 2010 and 2019, as well as between 2000 and 2010, which recorded a loss of 90.1 ha yr^{-1}. Major losses were observed close to human habitation in Northern Central swamps, Northern swamps, and Pate Island swamps. Both legal and illegal harvesting activities mostly drive the loss.

With a rotation cycle of 20 years for mangroves, at least 839,334 stems (or approximately 42,000 scores) of mangroves could be removed annually from the mangroves of Lamu. A significant number of harvestable poles will be *Fito* size class (15,155 scores/yr) followed by *Pau* (12,307 scores/yr), *Mazio* (6,661 scores/yr), *Boriti* (3,058 scores/yr), *Nguzo* (2,424 scores/yr), and *Mazio* (2,341 scores/yr).

Given the multiple products and services provided by Lamu mangroves, a structured approach is necessary, particularly for the commercial extraction of poles. This harvest management plan aims to promote sustainable utilization of mangrove wood resources in Lamu County while protecting nature. The plan proposes selective harvesting of poles from designated mangrove forest areas over prescribed periods. Poles will be harvested in designated annual coupes over a 20-year cycle, focusing on key species like *Rhizophora mucronata* and *Ceriops tagal*. To ensure natural regeneration, mature seed-bearing trees will be preserved to guarantee regeneration with desirable tree species and form.

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Abbreviations and Acronyms

CFA	Community Forest Association
CGL	County Government of Lamu
DBH	Diameter at Breast Height
FAO	Food and Agriculture Organization of the United Nations
GoK	Government of Kenya
KEFRI	Kenya Forestry Research Institute
KFS	Kenya Forest Service
KFSGO	Kenya Forest Service General Order
KMFRI	Kenya Marine and Fisheries Research Institute
KWS	Kenya Wildlife Service
LAMACOFA	Lamu Mangrove Community Forest Association
MKUCOFA	Mkunumbi Community Forest Association
MPA	Marine Protected Area
NMEMP	National Mangrove Ecosystem Management Plan
PANDAWE CFA	Pate- Ndau- Awer Community Forest Association
PFM **SCM**	Participatory Forest Management Supply Chain Management
SFM	Sustainable Forest Management
TNC	The Nature Conservancy
UNEP	United Nations Environment Programme

Glossary

Allowable Cut: Represents the quantity of wood or resources that can be sustainably extracted from a forest annually or within a management cycle. For mangroves, allowable cut calculations consider species recovery rates, growth under tidal influences, and ecological resilience (FAO, 1985: Manual of Forest Inventory Techniques Applicable to Mangroves).

Butt Diameter: The measurement of a tree's diameter near the base. This may involve measurements above aerial roots or pneumatophores in mangroves, depending on species characteristics (FAO, 1994). The butt diameter is crucial for volume estimation.

Degradation: Degradation in mangroves is defined as the decline in ecosystem services caused by over harvesting, hydrological alterations, or pollution, leading to reduced productivity and biodiversity loss (FAO, 2007; IPCC, 2019).

Ecosystem: A functional unit of interacting organisms and their environment. These include trees, water systems, tidal dynamics, and associated flora and fauna in mangroves, providing essential services such as shoreline protection and nutrient cycling (FAO, 1994).

Felling Plan: Refers to systematically divided forest areas designated for harvesting operations over a rotation period. This ensures continuity of forest cover and facilitates regeneration, particularly in sensitive mangrove ecosystems where staggered cutting can help maintain hydrological and ecological functions (FAO, 1994).

Forest: Land spanning more than 0.5 hectares with trees of at least 2 meters and a minimum canopy cover of 15%, including natural and plantation forests on state, community, and private land.

Harvest management Plan: A comprehensive strategy detailing the methods, timing, and areas for extracting forest resources, ensuring minimal environmental impact and alignment with sustainable management objectives. For mangrove forests, such plans often address tidal cycles, species-specific felling techniques, and measures to preserve adjacent ecosystems (FAO, 1994: Mangrove Forest Management Guidelines).

Merchantable Mangrove Species: Commercially valuable mangroves harvested for timber, poles, fuelwood, and other uses. In Kenya, *Rhizophora mucronata* and *Ceriops tagal* are the most preferred species due to their strong, durable wood, making them ideal for construction, boat building, and charcoal production. The commercial viability of these species depends on availability, sustainability, and legal regulations.

Merchantable stems: Refer to mangrove stems that are of good quality (Form 1 and 2) for construction purposes; all mangrove stems are categorized into three quality

classes (or Forms) based on the suitability of the lead stem for construction. Form 1 trees have straight poles with a ready market for construction, Form 2 trees have crooked poles requiring minor repair before being used for construction, and Form 3 trees have crooked poles unsuitable for building (Kairo et al., 2002).

Rotation Age: The period it takes a mangrove tree to grow to a desired size.

Utilization Zone: An area in a forest designated for resource extraction, often managed to balance ecological health and community needs. In mangrove regions, these zones are crucial for the sustainable harvesting of wood, honey, and aquaculture resources (FAO, 2007: The World's Mangroves 1980-2005).

CHAPTER 1 INTRODUCTION

1.1 Background Information

Historically, mangrove wood products from Lamu County were exploited for the export market in Arab countries. Records indicate that by the beginning of the 20th century, Kenya was exporting an annual average of 24,150 scores of mangrove poles, equivalent to 483,000 poles per year from Lamu forests. This export market its peak between 1947 and 1956 whereby a total of 75,000 scores whereby exported per year. Increased demand for poles exacerbated enormous pressure on the mangrove forests and led to a presidential ban on further export of mangrove wood from Kenya in 1982. Even with the export ban, domestic utilization of mangrove wood resources grew considerably over the years.

Obtain an archive photo

Currently, commercial harvesting of mangrove wood products remains a vital livelihood for over 30,000 households in Lamu, with the poles being sold in coastal urban centers for building and construction. Illegal harvesting of mangroves is perceived as the major threat facing mangroves in Lamu County. To address this, among other challenges, the National Mangrove Ecosystem Management Plan (2017-2027) prioritised the development of a Harvest Management Plan to guide mangrove exploitation in Lamu.

1.2 Plan Title and Duration

The plan shall be referred to as the **Mangrove Harvest Management Plan for Lamu County**. It shall be implemented within ten years, commencing from 2025 to 2035. The plan will be reviewed after five (5) years and revised after ten (10) years from the approval date.

1.3 Plan Purpose

The purpose of this mangrove harvest management plan is to:

1. Adhere to the Forest Conservation and Management Act (2016), which requires all forests in Kenya to be managed through an approved forest management plan.

2. Provide harvesting prescriptions and operations to be undertaken during the plan period.

3. Achieve a sustained annual supply of mangrove wood products to the market by providing off-take quantities of various mangrove products and their diameter classes.

4. Optimise the royalties earned by KFS from the station.

5. Outline procedures for annual operations and monitoring the Harvest management plan's implementation in line with the KFS work plans and reporting.

CHAPTER 2 MANGROVES OF LAMU COUNTY

2.1 Geographic Location

The mangroves of Lamu lie between latitudes 1° 40′ and 20° 30′ South and longitudes 40° 15′ and 40° 38′ East. They consist of a vast mainland and 65 islands forming the Lamu Archipelago, which has five major islands, i.e. Lamu, Manda, Pate, Kiwayu, and Ndau.

There are about 37,350 ha of mangroves in Lamu County, representing about 61 % of the total mangroves in Kenya. These forests are dominated by mixed stands of *Rhizophora mucronata* that account for 8,649 ha (or 23% of Lamu mangroves). Other notable mangrove forest formations include pure stands of *Avicennia marina* on the landward sides and *Ceriops tagal* in the mid-zone of the forest (Table 1).

Table 1: Areal extent of mangrove formation in Lamu (adapted from the National Mangrove Ecosystem Management Plan (2017-2027)

Classification	Area (ha)	% cover
Avicennia marina	6,966	18.7
Avicennia marina mixed with *Ceriops tagal*	1,961	5.3
Ceriops tagal	5,155	13.8
Mixture of *Ceriops tagal*, *Bruguiera gymnorrhiza, Rhizophora mucronata,* and *Avicennia marina*	1,901	5.1
Ceriops tagal-Rhizophora mucronata mix	5,138	13.6
Rhizophora mucronata	5,558	14.9
Mixture of *Rhizophora mucronata, Ceriops tagal, Bruguiera gymnorhiza, Avicennia marina*	8,649	23.2
Sonneratia alba	1,165	3.1
Sonneratia alba-Rhizophora mucronata mix	856	2.3
Total mangrove cover	**37,350**	**100**

2.2 Major Mangrove Areas in Lamu

For management purposes, mangroves in Lamu are classified into five management blocks, including the Northern swamps, Northern Central swamps, Mongoni-Dodori Creek swamps, Pate Islands swamps, and Southern swamps (Figure 1). Northern Central swamps account for 34% of the total mangroves in Lamu, followed by the Pate Island swamps (26%), Mongoni-Dodori Creek swamps (17%), Southern swamps (14%), and Northern swamps (8%) (Table 2). Mangroves occurring in the Kiunga Marine National Reserve (KMNR) include those in the Northern swamps and parts of the Northern Central swamps, with a total forest area of 7,628 ha; equivalent to 20.4% of the total mangrove coverage in Lamu County.

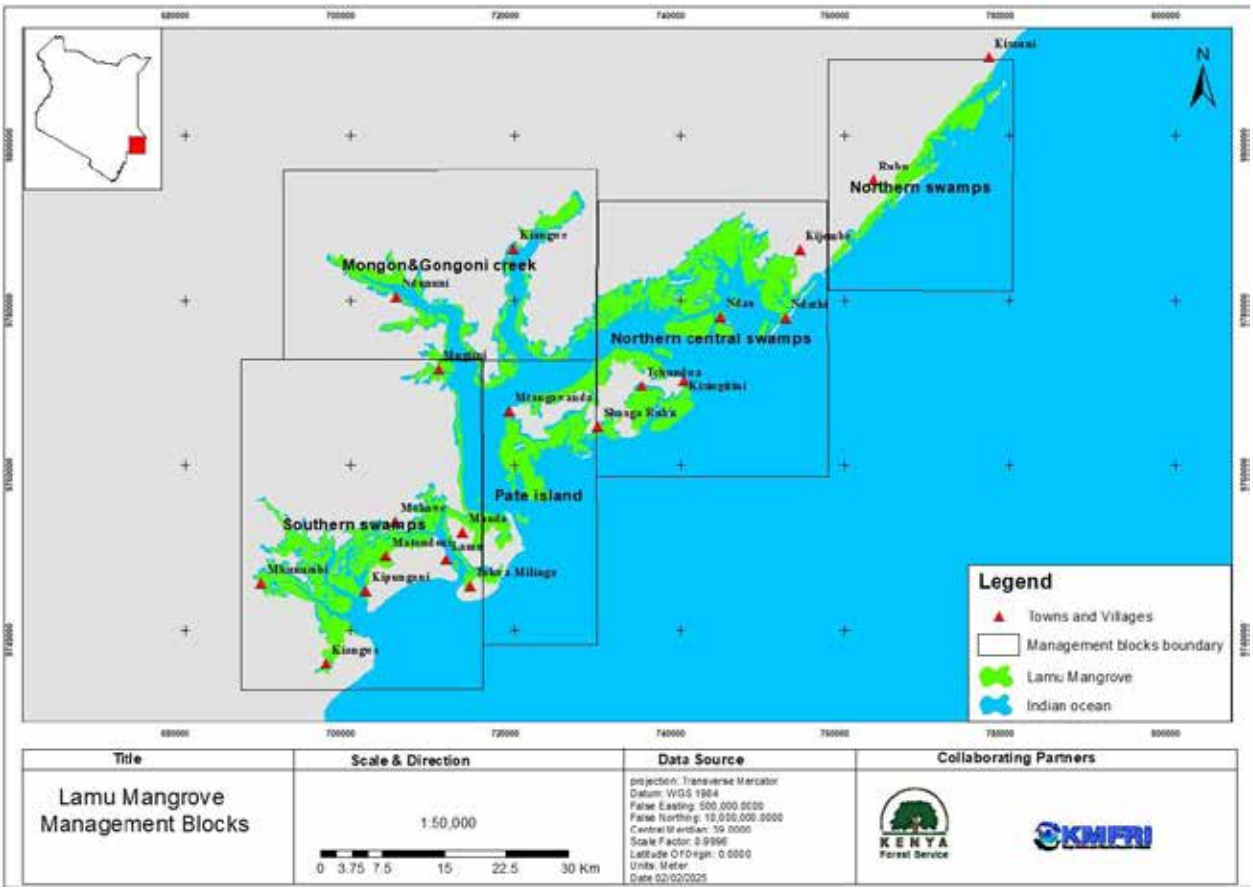

Figure 1: Distribution of mangrove forests within the five management blocks of Lamu County

Table 2: Description of the five mangrove management blocks of Lamu County and their respective areal coverage (values in parenthesis are % proportion cover of each block)

Management block	Description	Mangrove cover (ha)
Northern swamps	Stretch from Kiunga to Mlango wa Chano and is characterized by extensive *Rhizophora sp.* -dominated pure stands.	3,160 (8%)
Northern Central swamps	It extends from Mlango wa Chano in Kiunga to the mouth of Dodori Creek, including Uvondo and Ndau Islands. The block is highly stocked with *Ceriops sp.* and *Rhizophora sp.*	12,850 (34%)
Mongoni and Dodori Creek swamps	It encompasses mangroves along the shores of Mongoni-Dodori Creek and Ndununi areas, predominantly featuring stands of *Ceriops sp.*	6,400 (17%)
Pate Island swamps	Incorporates mangroves surrounding Pate Island, Shindabwe, Kizingitini, and Chongoni.	9,740 (26%)
Southern swamps	Comprises the mangroves of Kililana, Manda, Shela, Amu, Matondoni, Bandari Salama, Mokowe, Mkunumbi and Kimbo creeks.	5,200 (14%)

Over the years, mangroves in Lamu have suffered significant degradation punctuated by a few restoration efforts. About 1,739 ha (4.87%) of mangroves were lost between 1990 and 2019, translating to a general annual decline of 60 ha (0.16%). The losses

were perpetrated by the over-harvesting of wood products, conversion pressure and climate change. The highest decline rate was between 2010 and 2019, which recorded a loss of 114 $hayr^{-1}$, followed by the period between 2000 and 2010, which lost 90.1 $hayr^{-1}$. Significant losses happened in areas close to human habitation, mainly in the Northern Central swamps, Northern swamps, and Pate Island swamps.

In this harvest management plan, harvesting operations will be restricted to the designated utilization zone (10,766 ha) as prescribed in the national mangrove ecosystem management plan (2017-2027), excluding the proposed blue carbon project site, conservation, protection, and development zones (Figure 2). The highest proportion of utilization zones in Lamu is in the Northern Central swamps, at 5,412 ha, followed by Mongoni-Dodori swamps, Southern swamps, and Pate Island swamps (Table 3).

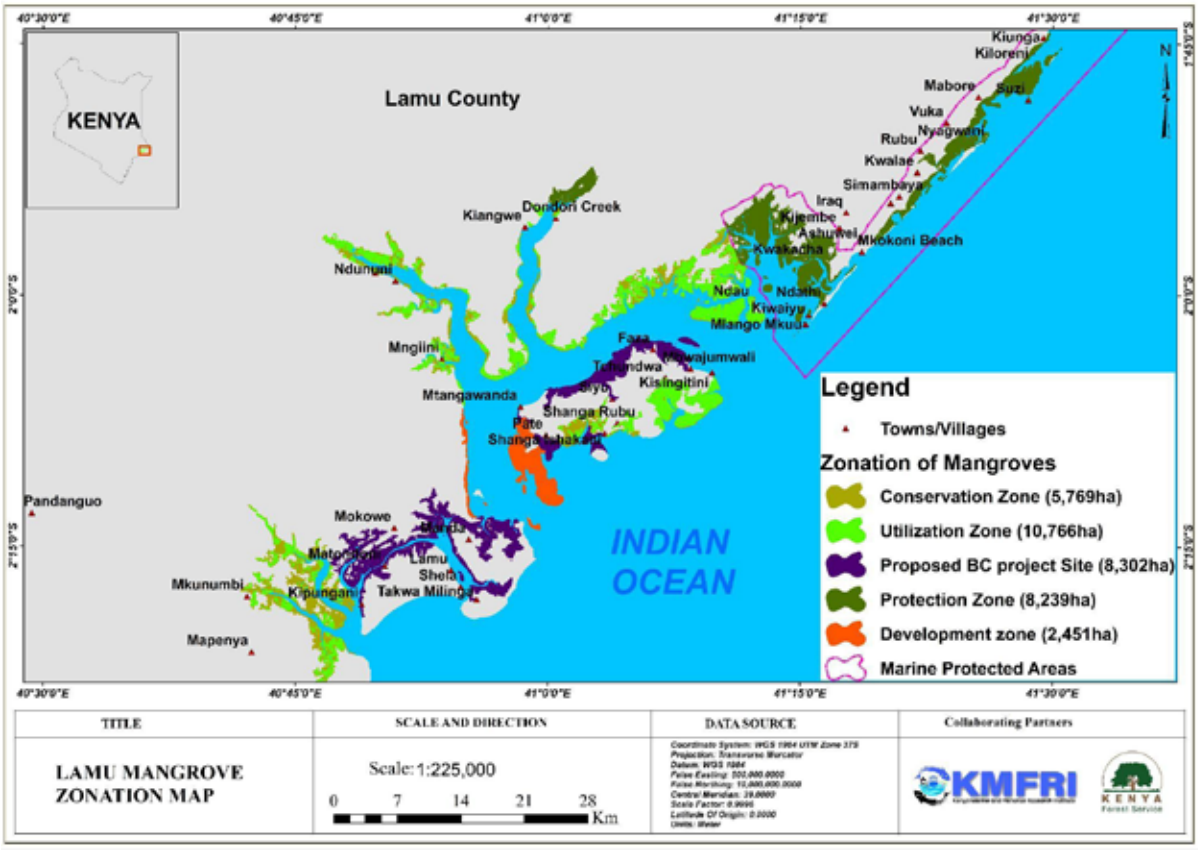

Figure 2: A map of Lamu mangroves showing different utilization zones

Table 3: Utilization zones of mangrove in Lamu

Management block	Utilization zones (Areas)	Area (ha)	Stocks of quality (merchantable) stems	Scores
Northern Central swamps	Ndau, Mbwajumwali, Tchundwa,Shanga Ishakani, Shanga rubu, Vumbe, Sadani, Kiweni	5,412	5,888,460	294,423
Mongoni-Dodori creek swamps	Ngiini, Ndununi, Kiangwe	2,577	8,253,320	412,666
Pate Island swamps	Pate, Mtangawanda	675	648,000	32,400
Southern swamps	Kiongwe, Mkunumbi, Hidio, Milihoi	2,102	1,996,891	99,845

2.3 Mangrove Utilization Classes

The utilization classes with butt diameter specifications are given in Table 4. The most preferred size classes are the *Boriti*-sized poles (9.1cm to13.0cm) used for construction, followed by *Mazio, Pau*, *Fito,* and *Nguzo*-sized poles in that order. Beyond the diameter of *Banaa (*20.1cm to 35cm*),* there is very little demand in the market.

Table 4: Description of different classes of mangrove wood products in Kenya

Utilization class	Diameter range (cm)	Length (m)	Unit of trade
Fito	<4	<4	score
Pau	4.1-6.0	>4	score
Mazio	6.1-9.0	>4	score
Boriti	9.1-13.0	>4	score
Nguzo	13.1-20.0	>4	unit
Banaa	20.1-35	>4	unit

(1 score = 20 poles)

2.4 Management History of Mangrove in Lamu

The management of mangrove forests in Lamu has a complex history marked by sporadic oversight and unsustainable exploitation. Since their gazettement in 1932, the mangroves lacked clear harvest management plans with extraction quantities determined arbitrarily by forest officers. The greatest demand was for poles from *Rhizophora mucronata* and *Ceriops tagal.* In 1947, the colonial government regulated exploitation through private firm concessions. The first mangrove working plans of 1951 instituted a 20-year rotation cycle, which began at Mkunumbi and moved northwards. Forest surveys in 1967 highlighted deforestation issues of mangroves in Lamu, but no harvest management plan was developed from the data.

In 1981, the then Forest Department conducted a forest inventory that culminated

in developing the Lamu Mangrove harvest management Plan of 1984. This harvest management plan focused on forest operations in Mongoni-Dodori Creek swamps, Pate Island swamps, Northern Central swamps, Northern swamps, and later the Southern swamps. harvesting in the Southern swamps was delayed because the forests were clear-felled in 1950–1951 and had not fully recovered.

The approach to mangrove management has since evolved towards participatory and decentralized strategies. The Forest Conservation and Management Act (2016), promotes community involvement through the Participatory Forest Management (PFM) framework, engaging local communities in decision-making, conservation, and sustainable harvesting. The National Mangrove Ecosystem Management Plan 2017-2027(NMEMP) was developed to provide a roadmap towards the rehabilitation, conservation, and sustainable utilization of mangrove resources in Kenya for the 2017–2027 period. The plan provides priority actions for each of the coastal counties with mangroves. For Lamu, the plan prioritised the development of a harvest management plan to guide the removal of mangrove wood products and ensure the system's long-term viability.

2.5 Current Utilization Patterns of Mangroves in Lamu

Commercial harvesting of mangrove wood (primarily for poles) is a major livelihood activity in Lamu County. While KFS regulates this activity through a licensing system, the current practice is largely market-driven rather than resource-based. The sustainability of these resources is a concern, as localized overexploitation of desired size classes is witnessed in most sites. Figure 3 shows the utilization patterns of mangrove wood products in Lamu for the 1992–2024 period.

Overall, the *Boriti*-sized poles were the highest removed at 170,769 scores, followed by *Nguzo* (96,923 scores), *Mazio* (55,000 scores), and *Pau* (12,692 scores) (KFS Database, 2024). *Fito*-sized poles are not traded but only used locally to fill the walls of traditional houses. Over the 1992-2024 period, the harvested poles have been significantly lower than the allowable Harvest management (Figure 3). This may point to under-exploitation of the forest wood resources.

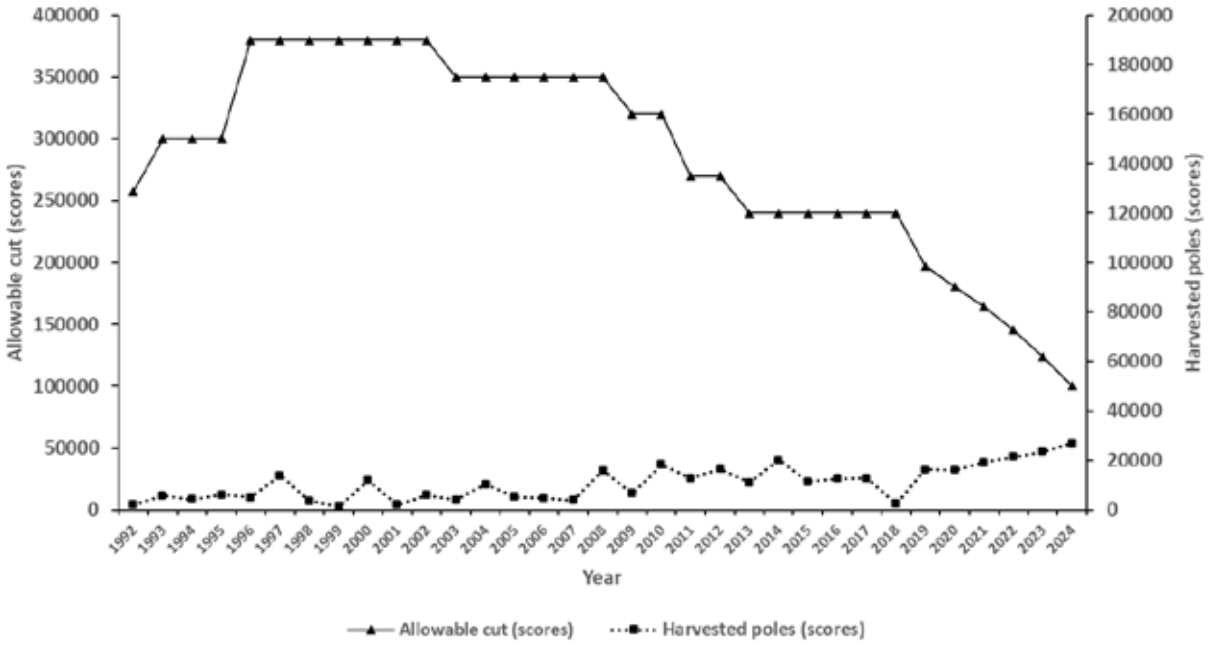

Figure 3: Mangrove poles harvesting (scores yr^{-1}) in Lamu, Kenya from 1992-2024; showing allowable and actual Harvest managements (1 score = 20 poles) (KFS database, 2024)

CHAPTER 3 MANGROVE FOREST STOCKS IN LAMU

3.1 Forest Inventory

The stocking levels used for this Harvest management plan were derived from the mangrove forest inventory conducted by KMFRI and KFS teams in 2019 in Lamu County and updated in January 2025. The data for the 2019 forest was derived from a total of 101 quadrats sampled in 64 transects. The transects were laid from the seaward-landward or channel bank inward across different mangrove types already marked out on Google Earth; the length of each transect depended on the locality and the extent of the forest zone. Vegetation sampling was conducted within 400 m² quadrats for mature trees and 25m² for natural regeneration, laid along the transects. Twenty-one (21) sampling plots were revisited in January 2025 to provide updated information and to consider pole extraction after lifting the 2018 ban on mangrove harvesting in Lamu County.

Within each quadrat, individual trees with butt diameters equal to or greater than 2.5cm were identified, counted, and their position mapped. Data on species, tree height, and DBH were measured from which basal area (m² ha^{-1}) and stand density (stems ha^{-1}) were estimated. Determination of Annual Allowable Cut (AAC) was based on FAO (1994)[1] guidelines following a modified Von Mantel's formula, which is entirely based on the growing (standing) stock:

$$AAC = 2\ (Gs/R)$$

Where,

AAC = Annual Allowable Cut (stems/scores)

Gs = Growing (standing) stock (stems/scores)

R = Rotation (years)

3.2 Current Stocking Levels of Mangroves in Lamu

All nine mangrove tree species have been observed in Lamu; however, only 8 species were represented during this inventory. Mangroves in Lamu are dominated by *Rhizophora sp.* and *Ceriops sp*, the most valuable species harvested for their wood products in Kenya. The stocking rates of *R. mucronata* and *C. tagal* range from 1,048–2,142 stems ha^{-1} (mean: 1425 stems ha^{-1}) and 104–967 stems ha^{-1} (mean: 605 stems ha^{-1}), respectively. Based on the 2019 mangrove inventory, the stand densities of mangrove trees in Lamu County are given in Table 5. This data was updated in January 2025 to focus on the utilization zones (Table 6).

1 FAO (1994). Mangrove Forest Management Guidelines. Rome: FAO Forestry Paper, 117, 319

Table 5: Stand density of mangrove forests in Lamu County

Management block	Area (ha)	Stocks		Quality (merchantable) stems		% of quality (merchant-able) stems
		Density (Stems ha^{-1})	Total stems	Density (poles ha^{-1})	Total stocks	
Northern swamps	3,160	1,607	5,078,120	939	2,967,240	58
Northern Central swamps	12,850	2,523	32,420,550	1,469	18,876,650	58
Mongoni-Dodori Creek swamps	6,400	2,169	13,881,600	1,271	8,134,400	59
Pate Island swamps	9,740	2,302	22,421,480	960	9,350,400	42
Southern swamps	5,200	3,092	16,078,400	1,743	9,063,600	56

The densities of merchantable *Rhizophora sp.* and *Ceriops sp.* in the management blocks where harvesting is allowed are provided in Table 6. The merchantable poles are higher across all the utilization zones, indicating sufficient supplies of desired wood products for construction.

Table 6: Stocking levels of the different size classes of *Rhizophora sp.* and *Ceriops sp.* mangroves within the management blocks

Block	Species	Category	Utilization stock classes (stems ha^{-1})					
			Fito	*Pau*	*Mazio*	*Boriti*	*Nguzo*	*Banaa*
Northern Central swamps	*Ceriops tagal (Mkandaa)*	Merchantable	200	181	67	19	8	0
		Non-merchantable	17	28	50	50	17	3
	Rhizophora mucronata	Merchantable	269	272	242	100	67	100
		Non-merchantable	22	94	56	22	36	6
Mongoni-Dodori creek swamps	*Ceriops tagal (Mkandaa)*	Merchantable	250	200	90	20	5	0
		Non-merchantable	30	5	10	5	0	0
	Rhizophora mucronata (Mkoko)	Merchantable	675	440	200	170	145	90
		Non-merchantable	55	175	120	70	20	15
Pate Island swamps	*Ceriops tagal (Mkandaa)*	Merchantable	31	23	0	2	0	0
		Non-merchantable	5	11	5	0	2	0
	Rhizophora mucronata (Mkoko)	Merchantable	183	186	134	80	81	94
		Non-merchantable	103	84	59	33	53	33

Block	Species	Category	Utilization stock classes (stems ha^{-1})					
			Fito	*Pau*	*Mazio*	*Boriti*	*Nguzo*	*Banaa*
Southern swamps	*Ceriops tagal* (Mkandaa)	Merchantable	239	93	11	7	0	0
		Non-merchantable	136	32	18	4	7	0
	Rhizophora mucronata (Mkoko)	Merchantable	121	129	61	14	11	61
		Non-merchantable	61	139	86	79	68	50

BOX: 1.

i. The *Fito* poles are on average around 7 years old.
ii. The *Pau* poles have an average age of 15 years.
iii. The *Mazio* poles have an average age of 24 years
iv. The *Boriti* poles have an average age of 32 years
v. *Nguzo* poles are obtained at 40 years
vi. *Banaa* poles are obtained at 48 years

3.3 Rotation age of Mangroves

In Kenya, permanent sample plots (PSPs) for monitoring the growth of mangroves did not exist until 2014 and 2021; when KFS and KMFRI established them in Lamu at Matondoni, Kilifi (Marereni), Mombasa (Tudor Creek), and Kwale County (Vanga, Sii Island, Upper Mwena, and Gazi Bay). However, drawing from experience from other countries where data on growth of mangroves has been documented, a rotation age of 60 years for Lamu mangroves has been used for this harvest management plan.

Losses occur before mangrove trees reach the desired size classes, mainly due to suppression by the canopy and normal mortality. Based on previous studies, it has been observed that 10% of *Fito* fail to reach *Pau,* 50% of *Pau* fail to reach *Mazio*, 30% of *Mazio* fail to reach *Boriti*, and 20% of *Boriti* do not reach *Nguzo* (Figure 4). These assumptions are based on studies carried out in other mangrove areas and can apply to the Lamu mangroves.

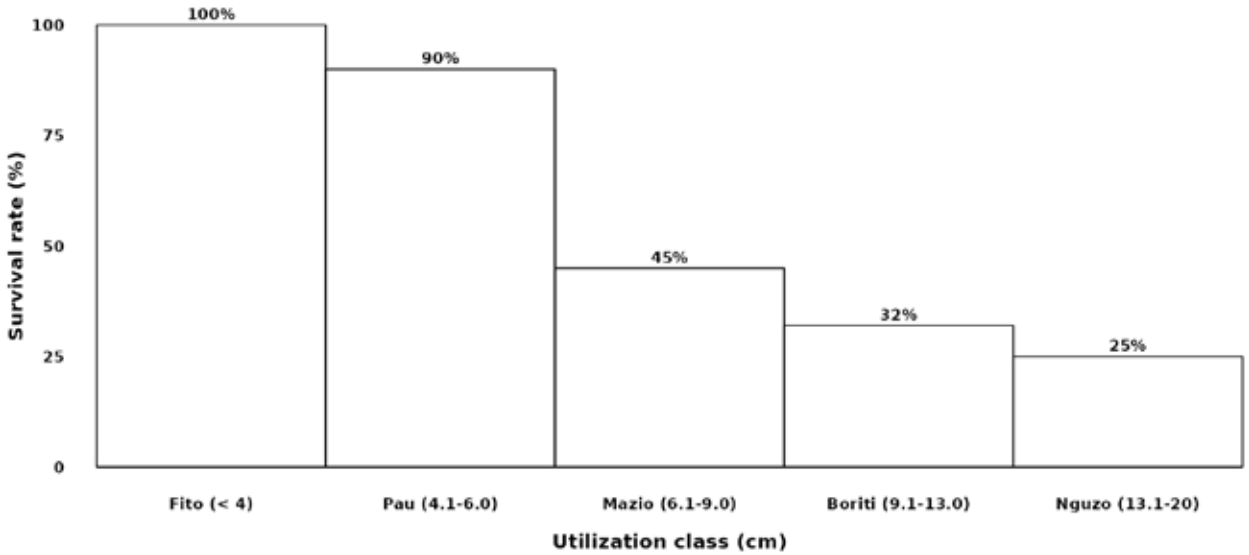

Figure 4: Recruitment (%) of mangroves into the next utilization class, progressing from *Fito* to *Nguzo*

3.4 Regeneration Stocking

To determine the regeneration stocking adequacy, the patterns and conditions of natural regeneration in the mangroves in the utilization zones were assessed. The saplings were categorised into three (3) classes: Regeneration Class I (RCI; all saplings of less than 40cm height), Regeneration Class II (RCII; all saplings of height ranging from 40cm to 150cm) and Regeneration Class III (RCIII: all saplings of over 150cm height but less than 2.5cm diameter). RCI represents potential regeneration, while RCII and RCIII represent established regeneration. Only the regeneration stocking of the mangrove species preferred for pole extraction (*Rhizophora sp.* and *Ceriops sp.*) was reported.

The Mongoni-Dodori creek swamps had the highest natural regeneration densities while the Northern Central block had the least; 7,450 ha^{-1} and 1,286 ha^{-1}, respectively (Table 7). In all the utilization zones, *Rhizophora sp.* had higher regeneration stocking than *Ceriops sp*. The regeneration ratios (RCI: RCII: RCIII) in all the blocks were less than the optimal levels, indicating that more restoration efforts need to be put in place. Assuming a mortality rate of 50% over a 10-year period, the projected regeneration stocking to be recruited into the merchantable stems is shown in Table 7.

Table 7: Natural regeneration densities (saplings ha^{-1}) of *Rhizophora sp.* and *Ceriops sp.* mangroves species in the utilization zones in Lamu. The last column indicates the projected regeneration stocking over 10 years, considering a mortality rate of 10% over the same period

Block	Species	Regeneration Stocking (Saplings ha^{-1})				
		RCI	RCII	RCIII	Total (2025)	Projection (2035)
Mongoni-Dodori creek swamps	*Ceriops tagal* (Mkandaa)	1,045	1,470	675	3,190	1,595
	Rhizophora mucronata (Mkoko)	1,175	1,800	1,285	4,260	2,130
	Total	**2,220**	**3,270**	**1,960**	**7,450**	**3,725**
	RCI: RCII: RCIII	1	2	1		
Northern Central swamps	*Ceriops tagal* (Mkandaa)	111	178	131	419	210
	Rhizophora mucronata (Mkoko)	494	81	292	867	433
	Total	**606**	**258**	**422**	**1,286**	**643**
	RCI: RCII: RCIII	1	1	1		
Southern swamps	*Ceriops tagal* (Mkandaa)	893	629	379	1,900	950
	Rhizophora mucronata (Mkoko)	311	1,196	986	2,493	1,246
	Total	**1,532**	**2,323**	**1,736**	**5,591**	**2,795**
	RCI: RCII: RCIII	**1**	**1**	**1**		

CHAPTER 4 HARVEST MANAGEMENT PLAN PRESCRIPTIONS

Mangrove harvesting is proposed to be carried out within the utilization zones i.e. Northern Central, Pate Island, Mongoni-Dodori creek, and Southern swamps, where commercial harvesting is allowed. Merchantable poles of all utilization classes have to be removed in one zone in one year to create evenness in later years. Thus, by the time all the present *Fito* have become *Nguzo,* there will be a form of evenness in the distribution of other size classes. The quantities estimated by first calculating the future crop based on growth timelines and growth rate of mangrove-size classes progressing from *Fito* to *Banaa* as shown in Figure 4 and Table 8.

Table 8: Growth rates of different mangrove utilization classes

Utilization class	Growth Rate (mm/year)	Duration (Years)	Optimum age
Fito	2.0-3.5	5-10	7
Pau	2.6-5.0	12-20	18
Mazio	3.0-3.9	22-30	28
Boriti	3.0-3.6	32-40	37
Nguzo	3.1	40-55	45
Banaa			

Source: Robert and Raura, 1967

Based on site characteristics, each swamp is given its own harvesting considerations. Therefore, future yield for each utilization zone was estimated for a period of ten (10) years with 2025 as the base year (Table 9).

Table 9: Future yield of mangrove poles from Lamu forests (Apart from *Nguzo* and

Banaa unit yield, all other utilization classes are in scores)

Management block	Utilization classes	Survival rates	Projected year and merchantable stems					
Northern Central swamps	Fito <4	90% (0.9)	**2025** 127,032					
	Pau 4.1-6.0 cm	50% (0.5)	**2033** 114,329	**2025** 122,522				
	Mazio 6.1-9.0 cm	70% (0.7)	**2042** 57,164	**2034** 61,261	**2025** 83,435			
	Boriti 9.1-13.0 cm	80% (0.8)	**2050** 40,015	**2042** 42,883	**2033** 58,405	**2025** 32,322		
	Nguzo 13.1-20.0 cm	90% (0.9)	**2058** 640,240	**2050** 686,121	**2041** 934,472	**2033** 517,147	**2025** 405,900	
	Banaa (20.1-35 cm)	-	**2066** 576,216	**2058** 617,509	**2049** 841,025	**2041** 465,432	**2033** 365,310	**2025** 541,200
Mongoni-Dodori creek swamps	Fito <4	90% (0.9)	**2025** 119,186					
	Pau 4.1-6.0 cm	50% (0.5)	**2033** 107,268	**2025** 82,464				
	Mazio 6.1-9.0 cm	70% (0.7)	**2042** 53,634	**2034** 41,232	**2025** 37,367			
	Boriti 9.1-13.0 cm	80% (0.8)	**2050** 37,544	**2042** 28,862	**2033** 26,157	**2025** 24,482		
	Nguzo 13.1-20.0 cm	90% (0.9)	**2058** 600,699	**2050** 461,798	**2041** 418,505	**2033** 391,704	**2025** 386,550	
	Banaa (20.1-35 cm)	-	**2066** 540,629	**2058** 415,619	**2049** 376,654	**2041** 352,534	**2033** 347,895	**2025** 231,930
Pate Island swamps	Fito <4	90% (0.9)	**2025** 8,640					
	Pau 4.1-6.0 cm	50% (0.5)	**2033** 7,776	**2025** 11,543				
	Mazio 6.1-9.0 cm	70% (0.7)	**2042** 3,888	**2034** 5,772	**2025** 2,869			
	Boriti 9.1-13.0 cm	80% (0.8)	**2050** 2,722	**2042** 4,040	**2033** 2,008	**2025** 1,485		
	Nguzo 13.1-20.0 cm	90% (0.9)	**2058** 43,546	**2050** 64,641	**2041** 32,133	**2033** 23,760	**2025** 148,500	
	Banaa (20.1-35 cm)	-	**2066** 39,191	**2058** 58,177	**2049** 28,920	**2041** 21,384	**2033** 133,650	**2025** 8,775

Management block	Utilization classes	Survival rates	Projected year and merchantable stems					
Southern swamps	Fito <4	90% (0.9)	**2025** 48,250					
	Pau 4.1-6.0 cm	50% (0.5)	**2033** 43,425	**2025** 29,619				
	Mazio 6.1-9.0 cm	70% (0.7)	**2042** 21,713	**2034** 14,810	**2025** 9,555			
	Boriti 9.1-13.0 cm	80% (0.8)	**2050** 15,199	**2042** 10,367	**2033** 6,688	**2025** 2,866		
	Nguzo 13.1-20.0 cm	90% (0.9)	**2058** 243,182	**2050** 165,867	**2041** 107,011	**2033** 45,862	**2025** 28,664	
	Banaa (20.1-35 cm)	-	**2066** 218,864	**2058** 149,280	**2049** 96,310	**2041** 41,276	**2033** 25,797	**2025** 162,427
Growth: Fito grows into Pau in 8 years, Pau grows to become Mazio in 9 years, Mazio grows to become Boriti in 8 years, Boriti grows to become Nguzo in 8 years, and Nguzo grows to become Banaa in 8 years.								

4.1 Working Cycle and Felling Plan

The working cycle was designed based on the size of the utilization zone in the blocks prescribed for pole harvesting and the historical exploitation within them. Therefore, the current cycle should start from Mongoni-Dodori creek swamps and proceed to the Northern Central swamps and then Southern swamps, with the Pate Island swamps being exploited in later years. The cutting cycle is shown in Table 10.

Table 10: Cutting cycle for mangroves of Lamu for the 2025 - 2035 period

Block	Coupe	Sites	Period/Year
Northern Central swamps	Ndau-Saadani	Ndau, Vumbe, Saadani	2026-2027
Northern Central and Mongoni-Dodori swamps	Saadani-Dodori	Saadani, Kiangwe, Dodori	2028-2029
Mongoni-Dodori creek swamps	Wange-Ndununi-Kwasasi	Wange, Ndununi, Ngiini, Kwasasi	2030-2031
Pate Island, Northern Central swamps	Chongoni-Rewa-Kizingitini	Chongoni, Pate, Rewa, Kizingitini	2032-2033
Southern Swamps	Mkunumbi	Mkunumbi, Hidiyo, Kiongwe, Mea, Kizuke, Ndambwe	2034-2035

The cutting procedure is indicated in Annex 1 for each type of pole: *Pau, Mazio, Boriti, Nguzo,* and *Banaa.* The same cutting pattern may be followed due to the high regeneration and high proportion of *Fito.* Using a decreasing method indicates how the cutting will continue progressively for each type. Higher cutting than the proposed scales will lead to degradation of the mangroves. Close monitoring of this cutting pattern should be done to assess its impact. The economic impact of the quantities to be cut each year is shown in the appendix. Potential wood products extractable from the mangrove of Lamu for the 2025-2035 period are given in Table 11.

Table 11. Lamu mangrove Annual Allowable Cut (AAC) in the four mangrove blocks for the 2025-2035 plan period

Year	Stocks	Scores (1 score = 20 poles/pieces)					Pieces		
		Fito	*Pau*	*Mazio*	*Boriti*	Total scores	*Nguzo*	*Banga*	Total Pieces
Mongoni-Dodori creek swamps	Available stock	119,186	82,464	37,367	24,482	**263,499**	386,550	231,930	**618,480**
	Allowable cut	11,200	8,246	3,737	2,448	**26,350**	38,655	23,193	**61,848**
Northern Central Swamps	Available stock	127,032	122,522	83,435	32,322	**365,311**	405,900	541,200	**947,100**
	Allowable cut	12,700	12,200	8,300	3,200	**36,400**	40,500	54,100	**94,600**
Southern swamps	Available stock	48,250	29,619	9,555	2,866	**90,290**	28,664	162,427	**191,091**
	Allowable cut	4,800	2,900	900	200	**8,800**	2,800	16,200	**19,000**
Pate Island swamps	Available stock	8,640	11,543	2,869	1,485	**24,537**	148,500	8,775	**157,275**
	Allowable cut	800	1,100	200	100	**2,200**	14,800	800	**15,600**
Total Annual Allowable Cut		**29,500**	**24,446**	**13,137**	**5,948**	**73,750**	**96,755**	**94,293**	**191, 048**

4.2 Financial Analysis

In order to realise sustainable mangrove exploitation, there is a great need to improve the capacity of the KFS to manage the mangrove resources effectively. The support required will include 3 boats (1 inboard engine boat of 120 horse power and two outboard engine boats of 75 horse power), construction of two outposts, and deployment of additional Forest rangers and scouts for the two new outposts. In addition, the security team needs to be provided with adequate fuel for boat patrols, security equipment, food and rations for the rangers and spare parts for boat repairs and maintenance. These are initial costs to allow Lamu Forest Station to manage and protect the mangroves effectively and sustainably.

Table 12 below compares the revenues and costs of implementing the Harvest management plan. The revenue estimates on the planned harvesting of mangrove poles at different classes and other royalties are based on realized royalties during the previous 5-year period, assuming the same level of royalties. Cost estimates are based on management operations prescribed in this management plan. The priority investments are based on the infrastructure and equipment development requirements.

The financial analysis shows that the direct costs of mangrove forest management is KSh. 321,340,511. Priority investments for the station to improve its forest management and utilization were identified during planning meetings. The estimated projected revenue to be collected during the plan period is KSh. 2,423,400,000, from the sale of poles and licences. It is worth noting that the investment costs and anticipated revenue have not considered other benefits likely to be accrued from improved protection of the entire forest, leading to additionality of forest benefits, such as carbon sequestration, improved fish breeding grounds and the nature-based enterprises.

T**able 12:** Revenue and Cost estimates for the 10 years of mangrove Harvest management plan for Lamu County

Revenue for 10 years		**Cost estimate for 10 years**	
Item	**Revenue (Kshs)**	**Item**	**Cost (Kshs)**
Sale of poles (Fito, Pau, Mazio, Boriti, Nguzo, & Banaa)	2,423,200,000	Purchase of 1 Large Boat	3,500,000
Licenses	200,000	Purchase of 5 Small Boats	7,500,000
		Purchase of Fuel and Lubricants	53,568,000
		Purchase of two (2) administration vehicles	20,000,000
		Construction of New Outposts (Mtangawanda, Kiongwe, Kiunga, Mkokoni, and Matondoni)	30,000,000
		Deployment of 25 additional Forest Rangers	145,759,737

		Boat repairs and maintenance	15,000,000
		Procure additional Security equipment	16,800,000
		Administrative costs	29,212,774
Total	**2,423,400,000**	**Total**	**321,340,511**

4.3 Operational Planning, Monitoring and Evaluation

This Harvest management plan is operationalized through the KFS annual work planning and budgeting process, which results in the annual work programme for each forest station. The basis for formulating the annual work plan is the prescriptions outlined in the Harvest management plan and the calendar of events guiding the timing of each silvicultural activity.

Monitoring and evaluation of the implementation of the Harvest management plan is essential since it provides a basis for correction, adjustment, and improvement to the proposed goals, targeted activities, and assessment of the achievements. Strategies to address problems encountered in the implementation of planned activities will be outlined; notwithstanding, a reflection of the past is made in a bid to make the future successful.

Monitoring will be continuous, while evaluation will be mid-period and after ten years. During the implementation period, continuous monitoring will be through the preparation and submission of monthly, quarterly, and annual progress reports. Monitoring and evaluation will be undertaken in a consultative manner and will aim to assess the achievement of objectives, analyse and address constraints, thus providing information essential in the revision of the Harvest management plan.

KFS, alongside other collaborating agencies, will monitor and evaluate implementation of the plan as guided (Table 13). Field officers will be responsible for preparing and submitting monthly, quarterly, and annual progress reports.

Table 13: Monitoring and Evaluation Guide

Activity	**Indicator**	**Means of verification**	**Timeline**	**Responsible Office**
Sale of poles (*Fito, Pau, Mazio, Boriti, Nguzo, & Banaa*)	Amount of revenue collected No. of Movement permits issued	Valid list of prequalified licensees Payment records including invoices and receipts	Annual	RFC CFC FSM
Procurement of various goods, services and equipment	Quantity of goods, services and equipment delivered No. of delivery notes	Procurement plans Tenders, Request for quotations Invoices, payment receipts, Delivery notes	Annual	SCM RFC CFC FSM

Construction of New Outposts (Mtangawanda, Kiongwe, Kiunga, Mkokoni, and Matondoni)	No. of Outposts constructed Percentage of completion	Completion certificate available	Annual	SCM RFC CFC FSM
Repairs and Maintenance	No. of serviceable vehicles, motorcycles and boats	Receipts for payment for spares and service charge	Annual	RFC CFC FSM
Personnel deployment	Optimum number of personnel per outpost	Approved deployment schedule	Annual	RFC CFC FSM

CHAPTER 5 CONCLUSION

Mangroves are valuable resources to the country and particularly impact the people of Lamu because they are a main source of livelihood. The demand for mangrove wood products continues to increase due to population growth, hence there is need to invest in sustainable management and harvesting plans.

Fast-growing tree species to serve as an alternative wood source is challenged by the arid climate and the heavy capital investment needed. When managed well, mangroves shall continue to provide the desired wood products without affecting the productivity and resilience of the Mangrove Eco-system. Potential wood products extractable from the mangrove of Lamu for the 2025-2035 period are given in Table 11.

ANNEXES

Annex 1: *Fito* felling plan in scores

Year	Stocks	Mongoni-Dodori swamps	Northern Central swamps	Southern swamps	Pate Island swamps
2025	Available stock	119,186			
	Allowable cut	11,919			
2026	Available stock	107,268			
	Allowable cut	10,727			
2027	Available stock	96,541			
	Allowable cut	9,654			
2028	Available stock	86,887			
	Allowable cut	8,689			
	Balance	**78,198**			
2029	Available stock		127,032		
	Allowable cut		12,703		
2030	Available stock		114,329		
	Allowable cut		11,433		
2031	Available stock		102,896		
	Allowable cut		10,290		
2032	Available stock		92,606		
	Allowable cut		9,261		
	Balance		**83,345**		
2033	Available stock			48,250	
	Allowable cut			4,825	
2034	Available stock			43,425	
	Allowable cut			4,343	
	Balance			**39,083**	
2035	Available stock				8,640
	Allowable cut				864
	Balance				**7,776**

Annex 2: *Pau* felling plan in scores

Year	Stocks	Mongoni-Dodori swamps	Northern Central swamps	Southern swamps	Pate Island swamps
2025	Available stock	82,464			
	Allowable cut	8,246			
2026	Available stock	74,218			
	Allowable cut	7,422			
2027	Available stock	66,796			
	Allowable cut	6,680			
2028	Available stock	60,116			
	Allowable cut	6,012			
	Balance	**54,105**			
2029	Available stock		122,522		
	Allowable cut		12,252		
2030	Available stock		110,270		
	Allowable cut		11,027		
2031	Available stock		99,243		
	Allowable cut		9,924		
2032	Available stock		89,319		
	Allowable cut		8,932		
2033	Available stock		80,387		
	Allowable cut		8,039		
	Balance		**72,348**		
2034	Available stock			29,619	
	Allowable cut			2,962	
	Balance			**26,657**	
2035	Available stock				11,543
	Allowable cut				1,154
	Balance				**10,389**

Annex 3: *Mazio* felling plan in scores

Year	Stocks	Mongo-ni-Dodori swamps	Northern Central swamps	Southern swamps	Pate Island swamps
2025	Available stock	37,367			
	Allowable cut	3,737			
2026	Available stock	33,630			
	Allowable cut	3,363			
2027	Available stock	30,267			
	Allowable cut	3,027			
	Balance	**27,241**			
2028	Available stock		83,435		
	Allowable cut		8,344		
2029	Available stock		75,092		
	Allowable cut		7,509		
2030	Available stock		67,582		
	Allowable cut		6,758		
2031	Available stock		60,824		
	Allowable cut		6,082		
2032	Available stock		54,742		
	Allowable cut		5,474		
	Balance		**49,268**		
2033	Available stock			9,555	
	Allowable cut			956	
2034	Available stock			8,600	
	Allowable cut			860	
	Balance			**7,740**	
2035	Available stock				2,869
	Allowable cut				287
	Balance				**2,582**

Annex 4: *Boriti* felling plan in scores

Year	Stocks	Mongo-ni-Dodori swamps	Northern Central swamps	Southern swamps	Pate Island swamps
2025	Available stock	24,482			
	Allowable cut	2,448			
2026	Available stock	22,034			
	Allowable cut	2,203			
2027	Available stock	19,830			
	Allowable cut	1,983			
2028	Available stock	17,847			
	Allowable cut	1,785			
	Balance	**16,063**			
2029	Available stock		32,322		
	Allowable cut		3,232		
2030	Available stock		29,090		
	Allowable cut		2,909		
2031	Available stock		26,181		
	Allowable cut		2,618		
2032	Available stock		23,563		
	Allowable cut		2,356		
2033	Available stock		21,206		
	Allowable cut		2,121		
	Balance		**19,086**		
2034	Available stock			2,866	
	Allowable cut			287	
	Balance			**2,579**	
2035	Available stock				1,485
	Allowable cut				149
	Balance				**1,337**

Annex 5: *Nguzo* felling plan in stems

Year	Stocks	Mongo-ni-Dodori swamps	Northern Central swamps	Southern swamps	Pate Island swamps
2025	Available stock Allowable cut	386,550 38,655			
2026	Available stock Allowable cut	347,895 34,790			
2027	Available stock Allowable cut	313,106 31,311			
2028	Available stock Allowable cut **Balance**	281,795 28,179 **253,615**			
2029	Available stock Allowable cut		405,900 40,590		
2030	Available stock Allowable cut		365,310 36,531		
2031	Available stock Allowable cut		328,779 32,878		
2032	Available stock Allowable cut **Balance**		295,901 29,590 **266,311**		
2033	Available stock Allowable cut **Balance**			28,664 2,866 **25,798**	
2034	Available stock Allowable cut				148,500 14,850
2035	Available stock Allowable cut **Balance**				133,650 13,365 **120,285**

Annex 6: *Banaa* felling plan in stems

Year	Stocks	Mongo-ni-Dodori swamps	Northern Central swamps	Southern swamps	Pate Island swamps
2025	Available stock	231,930			
	Allowable cut	23,193			
2026	Available stock	208,737			
	Allowable cut	20,874			
	Balance	**187,863**			
2027	Available stock		541,200		
	Allowable cut		54,120		
2028	Available stock		487,080		
	Allowable cut		48,708		
2029	Available stock		438,372		
	Allowable cut		43,837		
2030	Available stock		394,535		
	Allowable cut		39,453		
2031	Available stock		355,081		
	Allowable cut		35,508		
2032	Available stock		319,573		
	Allowable cut		31,957		
	Balance		**287,616**		
2033	Available stock			162,427	
	Allowable cut			16,243	
2034	Available stock			146,184	
	Allowable cut			14,618	
	Balance			**131,566**	
2035	Available stock				8,775
	Allowable cut				878
	Balance				**7,898**

Pole Size	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**
Fito	MD	MD	MD	MD	NC	NC	NC	NC	SS	SS	PIS
Pau	MD	MD	MD	MD	NCS	NCS	NCS	NCS	NCS	SS	PIS
Mazio	MD	MD	MD	NCS	NCS	NCS	NCS	NCS	SS	SS	PIS
Boriti	MD	MD	MD	MD	NCS	NCS	NCS	NCS	NCS	SS	PIS
Nguzo	MD	MD	MD	MD	NCS	NCS	NCS	NCS	SS	PIS	PIS
Banaa	MD	MD	NC	NC	NC	NC	NC	NC	SS	SS	PIS

Where, MD stands for Mongoni-Dodori, PIS stands for Pate Island swamps, NC stands for Northern Central swamps, SS stands for Southern swamps, and NS stands for Northern swamps.